Practitioner-oriented studies

Introduction to the topic: industries, stakeholders, processes, guidance, case studies

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Intro to this part of the course

- Rationale: you have seen many examples and covered key concepts in transport modelling. This topic is about bringing this knowledge together and seeing how it can work in practice. This means that we cannot study models in isolation, but we must consider the entire life cycle of transport-related projects, from inception, through methods/model development, to reporting.

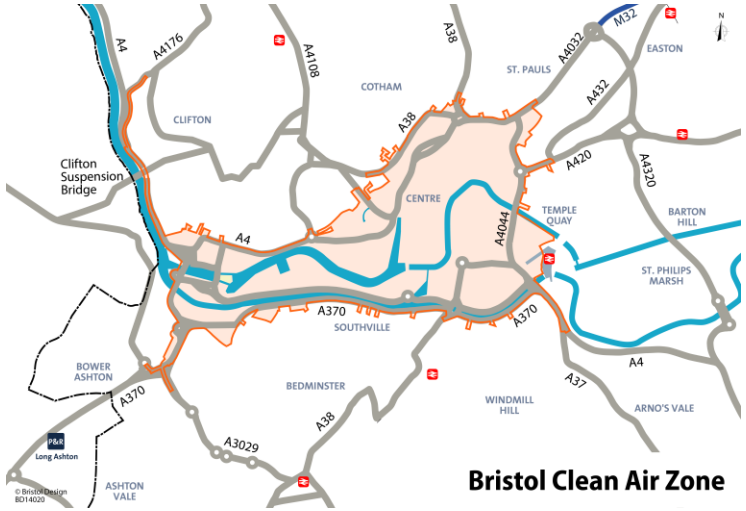
- This part of the course includes:
  1. Lecture notes/reading
     i. An introduction (these slides)
     ii. Case studies (provided separately on Blackboard)
  2. Discussion of case studies (Live session)
  3. An end-to-end example of a consultancy project (end of Live session and Lab)

Outline

These slides will cover introductory examples for:

- Industries employing transportation modelling
- How/where transport modelling fits into design/decision processes
- Transport modelling guidance
- Commercial transport models
- Case studies

Industries





Examples:

- Local and central government
- Public transport providers
- Building managers, e.g. football stadiums
- Event managers, e.g. Oktoberfest in Munich
- Transport consultants





Process example: consultancy project

Examples:

- Safety checks for events
- Circulation of visitors in museums
- Evacuation analysis for buildings

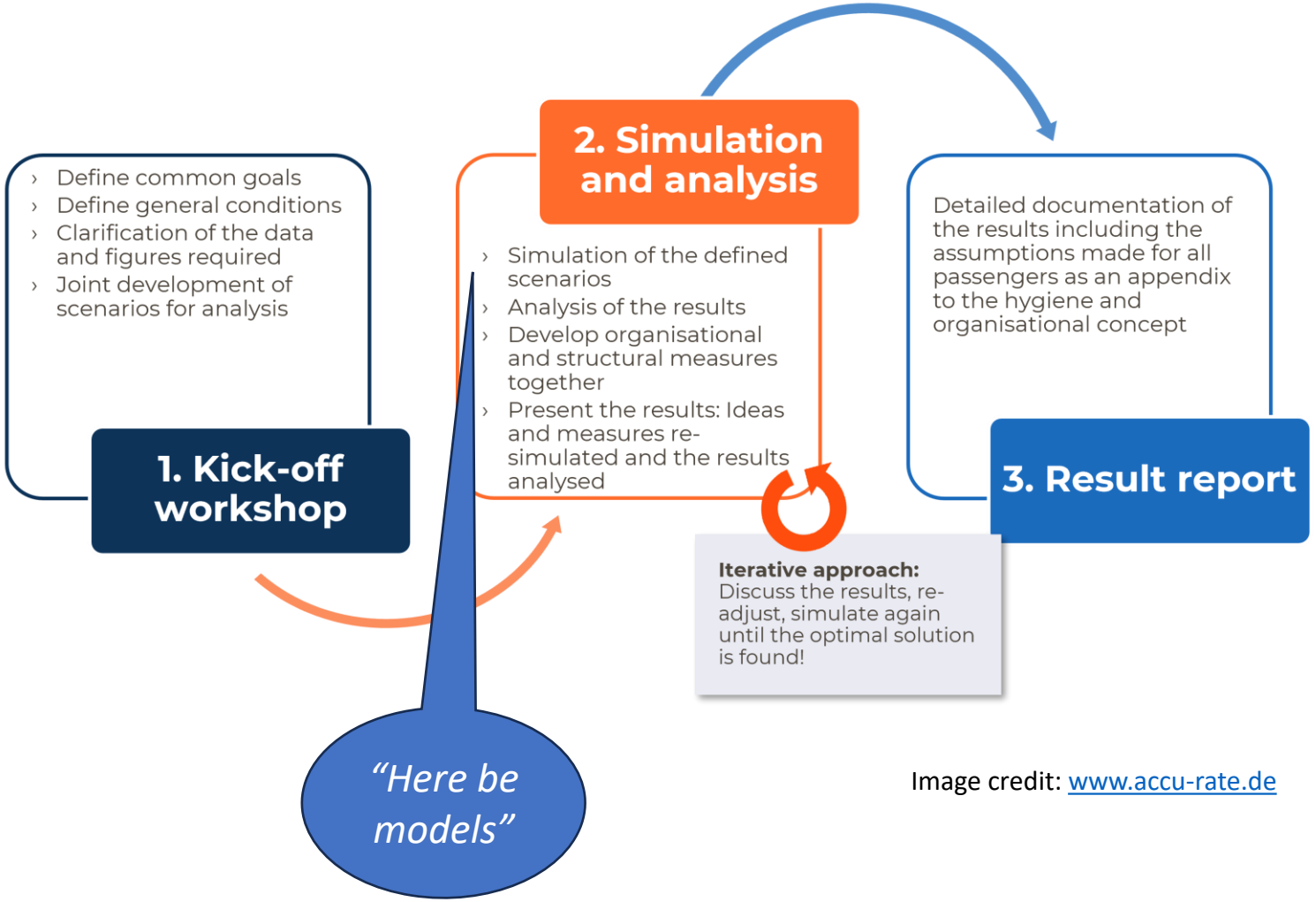


Image credit: www.accu-rate.de

Process example: department for transport

Examples:

- High Speed 2
- Bristol Clean Air Zone
- …

Department for Transport

Guidance

Transport business case guidance

Updated 16 December 2022

Contents

Foreword
Introduction
The business case
The strategic dimension
The economic dimension
The financial dimension
The commercial dimension
The management dimension
Tools and guidance

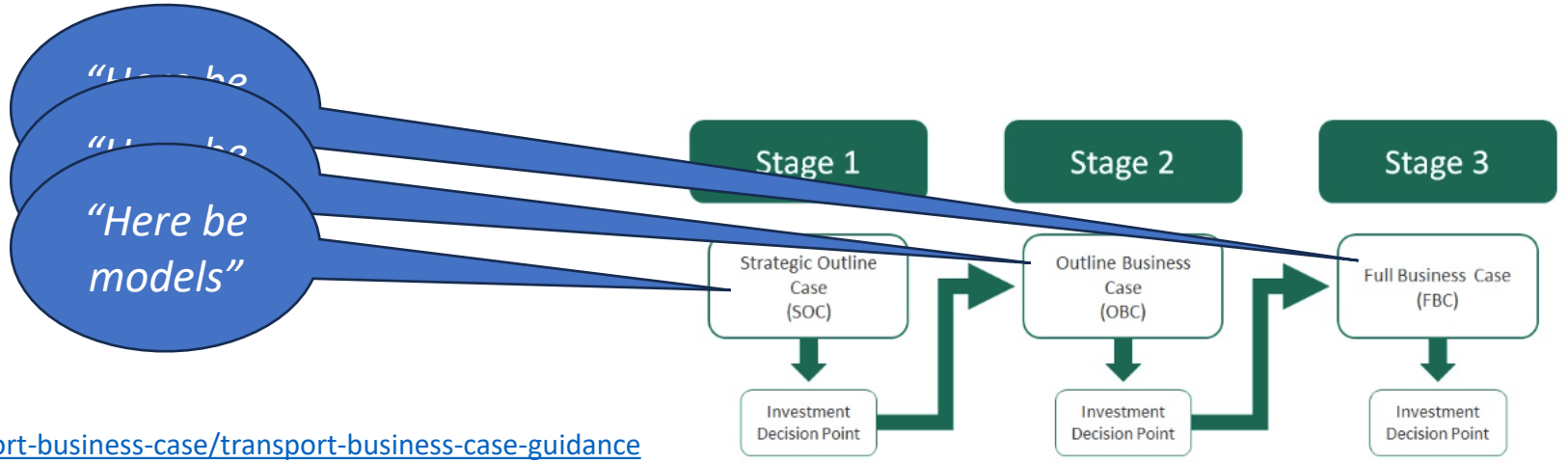
Print this page

Overview of the process for a business case

The programme business case should be developed iteratively as one document, whereas the project business case is typically developed in 3 stages. Smaller, more straightforward policies or investment proposals may require fewer phases while larger or complex proposals may require more stages.

However, all of the key steps and actions should be covered and the reduction in stages must be agreed in advance with the reviewing authority. The 3 stages are:

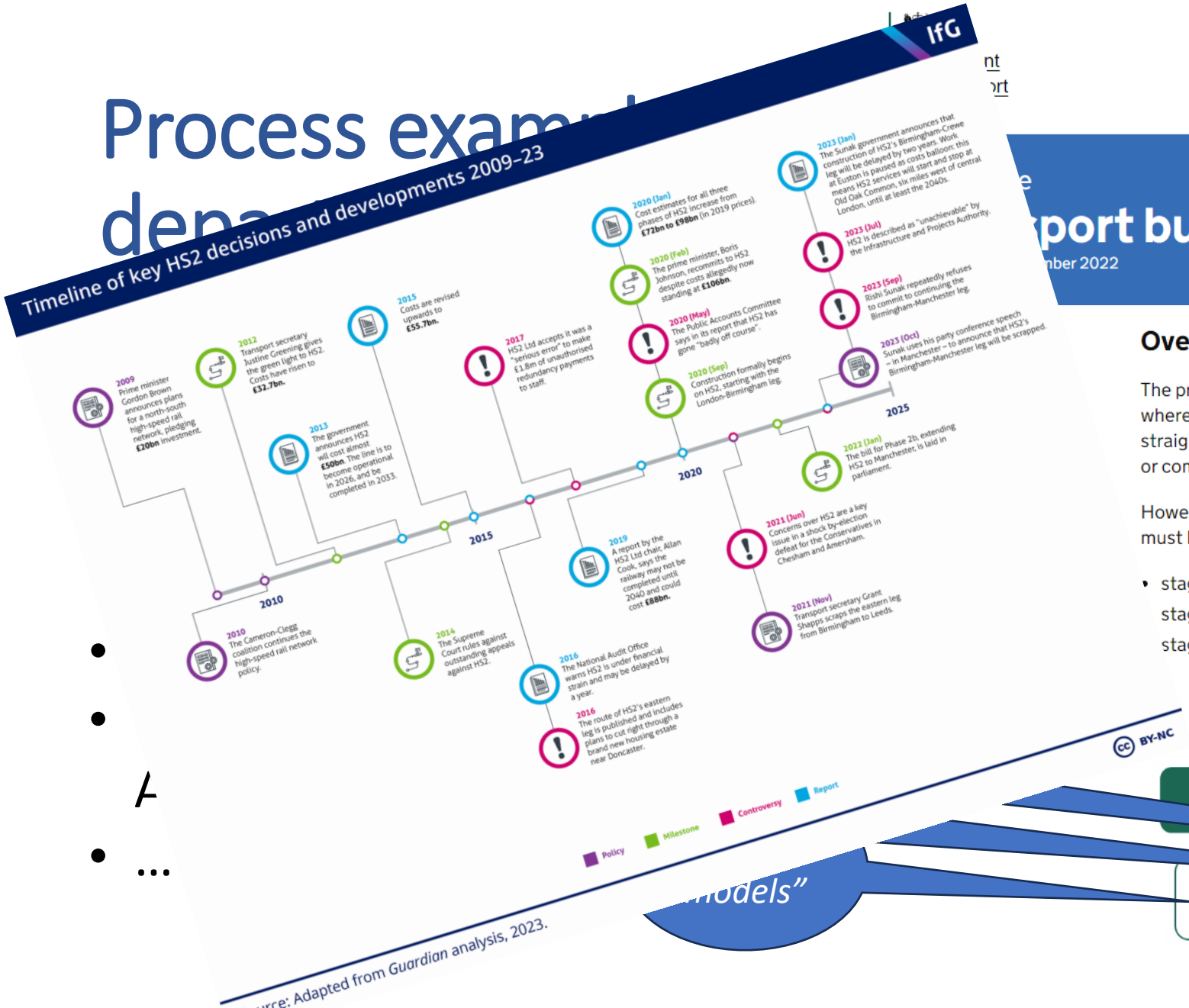
- stage 1 – strategic outline case (SOC)
- stage 2 – outline business case (OBC)
- stage 3 – full business case (FBC)



Source: https://www.gov.uk/government/publications/transport-business-case/transport-business-case-guidance

Process exam… dep…

- …
- …
- …



"…models"

BY-NC

…port business case guidance

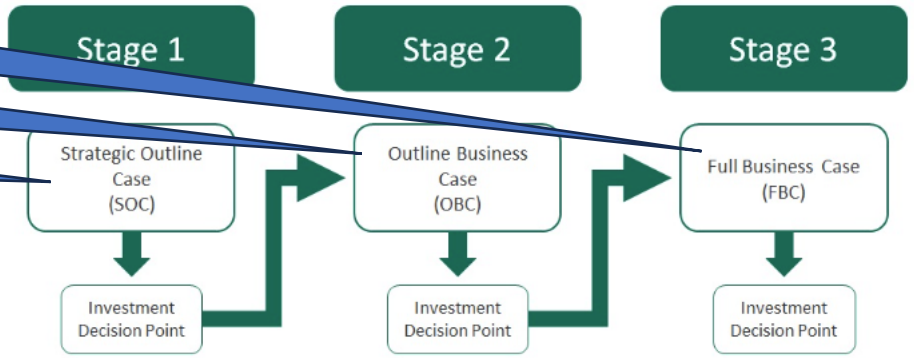
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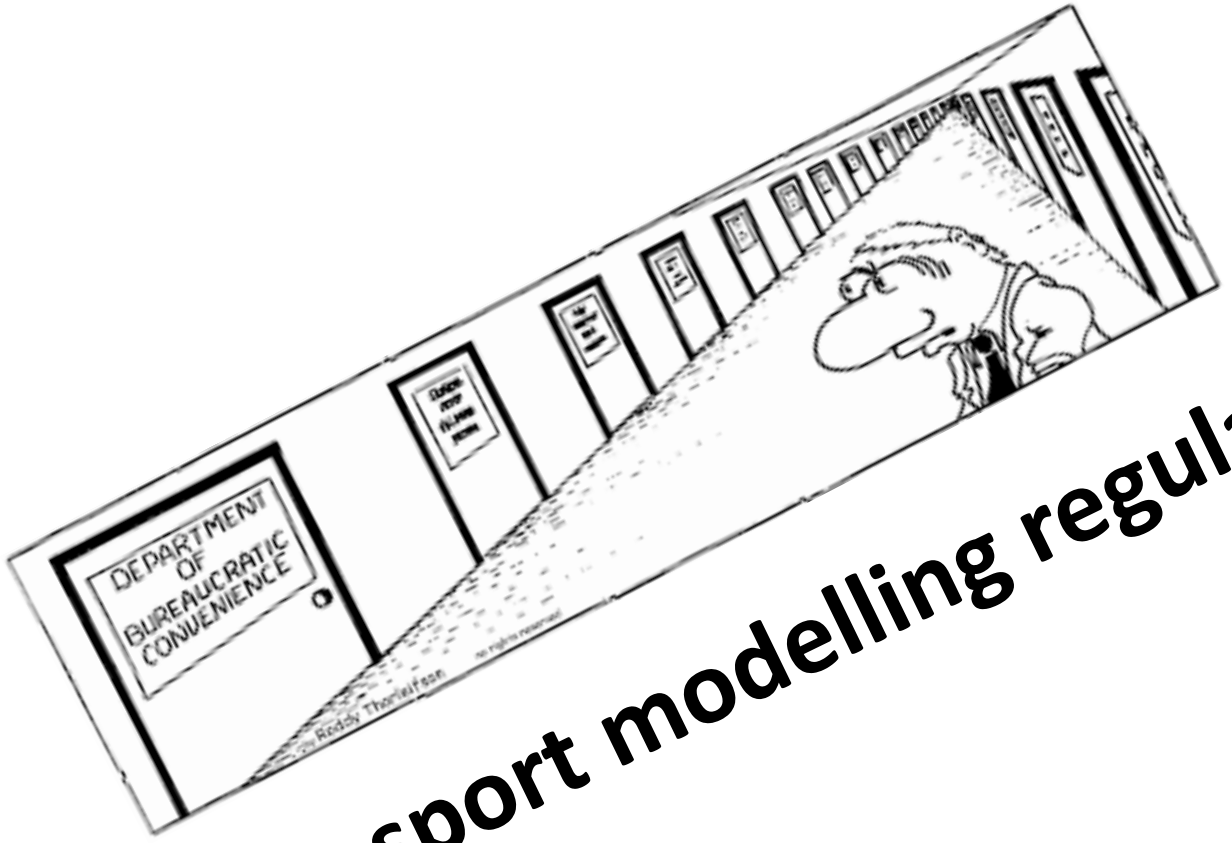
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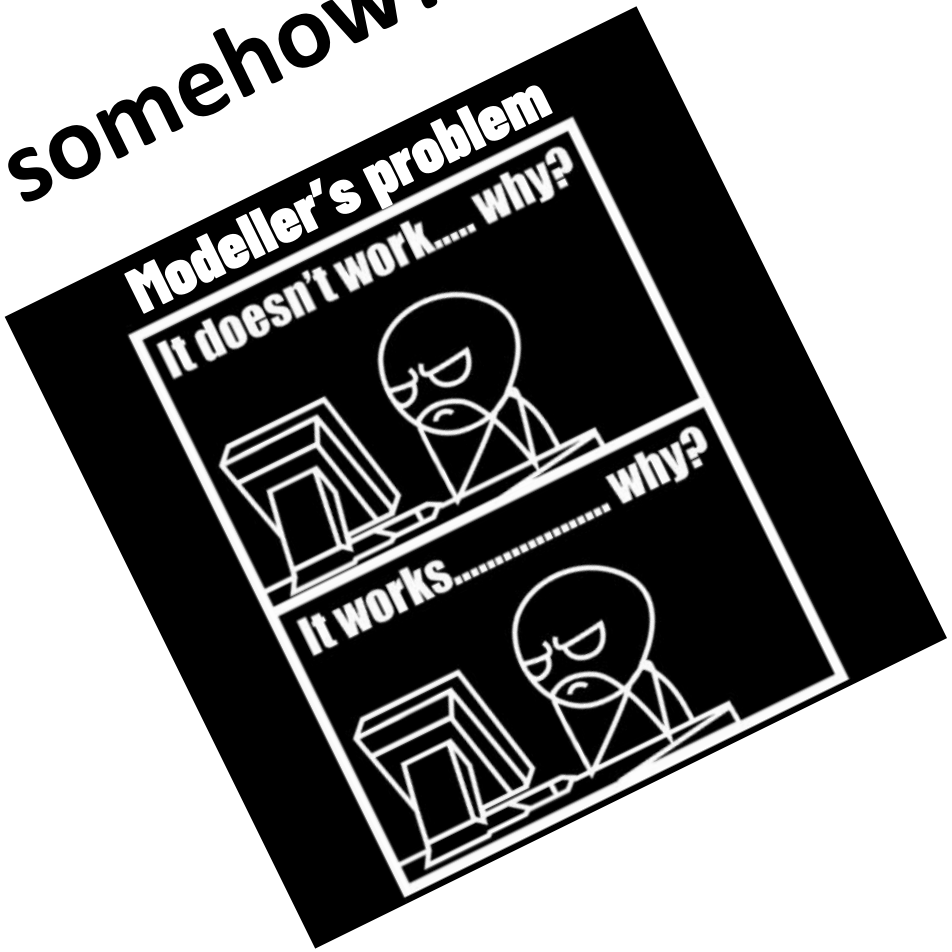


Source: https://www.gov.uk/government/publications/transport-business-case/transport-business-case-guidance

Transport modelling guidance?



Is transport modelling regulated somehow?



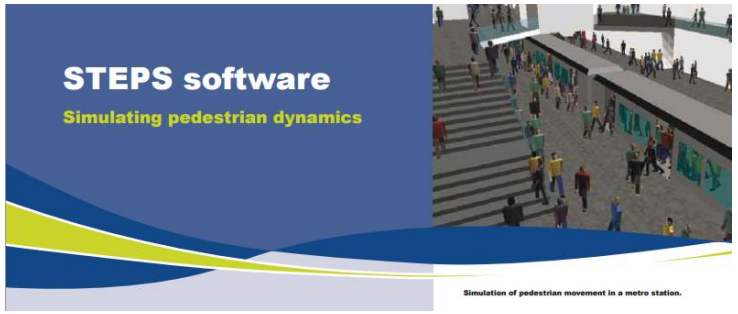
Examples for transport modelling guidance

- Transport analysis guidance (TAG)
  - Information on transport modelling and appraisal
  - Specifies demand, assignment, forecasting, advanced modelling guidance
  - Provides links to software tools (e.g. SATURN assignment model)
  - https://www.gov.uk/guidance/transport-analysis-guidance-tag#guidance-for-the-modelling-practitioner

- DIN 18009-2 (Germany)
  - Evacuation simulation standard
  - see e.g.: https://doi.org/10.17815/CD.2024.152

- Transport modelling guidance (Transport for London)
  - Comprehensive, covers all transport modes
  - https://tfl.gov.uk/corporate/publications-and-reports/transport-modelling-guidance

Examples for commercial transport models



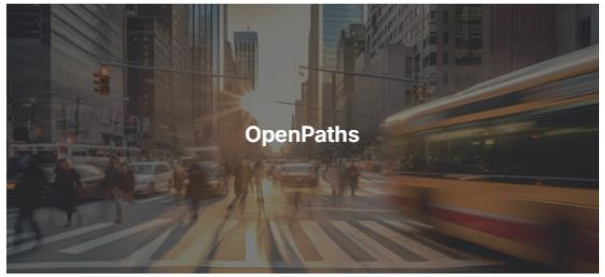
buildingEXODUS
the evacuation model for the building industry

maritimeEXODUS
the evacuation model for the maritime industry

airEXODUS
the evacuation model for aviation

SATURN
Congested Highway Assignment Software

Mobility Simulation Software



Multimodal, integrated transport planning, modeling, and analysis software that opens paths to better mobility.



Model & simulate the movement of people to optimize pedestrian infrastructure

PTV VISSIM
the mind of movement

PTV VISWALK
the mind of movement

crowd:it
a product by accu:rate

… as a student, you can download lots of this software for free to try it out…

Case studies

- In this topic, we explore several case studies of real-world transport modelling applications.

- You can find documents and information about these case studies on Blackboard.

- When reading them, consider the following questions:
  1. How was modelling used? If it wasn't used, how could it have been used?
  2. What data was used? Can you think of 'better' data than what was used?
  3. Who are the key stakeholders, and how do they interact with the models that were being used?

Summary

We have seen:

- Industries employing transportation modelling
- How/where transport modelling fits into design/decision processes
- Transport modelling guidance
- Commercial transport models
- Case studies